

PRACTICAL - 1

```
#include <iostream>
```

```
#include <vector>
```

```
#include <chrono>
```

```
using namespace std;
```

```
using namespace chrono;
```

```
// Linear Search
```

```
int linearSearch(vector<int> arr, int key)
```

```
{
```

```
    int i = 0;
```

```
    while (i < arr.size())
```

```
    {
```

```
        if (arr[i] == key)
```

```
            return i;
```

```
        i++;
```

```
    }
```

```
    return -1;
```

```
}
```

```
// Binary Search
```

```
int binarySearch(vector<int> arr, int key)
```

```
{
```

```
    int first = 0;
```

```
    int last = arr.size() - 1;
```

```
while (first <= last)
{
    int middle = (first + last) / 2;

    if (arr[middle] == key)
        return middle;

    if (key > arr[middle])
        first = middle + 1;
    else
        last = middle - 1;
}

return -1;
}

int main()
{
    int n, key;

    cout << "Enter the size of the array: ";
    cin >> n;

    vector<int> arr(n);

    // Store sorted elements
    for (int i = 0; i < n; i++)
```

```

{
    arr[i] = i + 1;
}

cout << "Enter the element to search: ";
cin >> key;

if (key < 1 || key > n)
{
    cout << "\nElement is outside the array range.\n";
    return 0;
}

int position;

// Linear Search Timing
auto start1 = high_resolution_clock::now();
position = linearSearch(arr, key);
auto end1 = high_resolution_clock::now();

auto linearTime = duration_cast<microseconds>(end1 - start1);

cout << "\n===== Linear Search =====\n";
if (position != -1)
    cout << "Element found at index: " << position << endl;
else
    cout << "Element not found." << endl;

```

```
cout << "Execution Time: " << linearTime.count() << " microseconds\n";
```

```
// Binary Search Timing
```

```
auto start2 = high_resolution_clock::now();
```

```
position = binarySearch(arr, key);
```

```
auto end2 = high_resolution_clock::now();
```

```
auto binaryTime = duration_cast<microseconds>(end2 - start2);
```

```
cout << "\n===== Binary Search =====\n";
```

```
if (position != -1)
```

```
    cout << "Element found at index: " << position << endl;
```

```
else
```

```
    cout << "Element not found." << endl;
```

```
cout << "Execution Time: " << binaryTime.count() << " microseconds\n";
```

```
// Compare Times
```

```
cout << "\n===== Performance Comparison =====\n";
```

```
if (linearTime.count() > binaryTime.count())
```

```
    cout << "Binary Search is faster than Linear Search.\n";
```

```
else if (linearTime.count() < binaryTime.count())
```

```
    cout << "Linear Search is faster than Binary Search.\n";
```

```
else
```

```
    cout << "Both searches took the same time.\n";
```

```
return 0;
```

}